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The Fragility Index of Risk Factors for Hamstring Injuries

Hamstring injuries are one of the most prominent injuries in sports.[24] There has been an exponential increase in research assessing hamstring injury risk factors. Risk factor studies often employ a prospective cohort study design, where specific variables (eg, age, injury history, strength, flexibility) are studied in cohorts of participants (typically n>100) at the start of a (pre-)season. Individuals are classified as at either high risk or low risk of injury.

Hamstring injury is monitored across a specified period (eg, the remainder of the season), with participants ultimately being classified as either sustaining an injury or not. From this information, a 2 × 2 contingency table can be developed, with participants classified as either (1) high-risk and sustained a hamstring injury, (2) high-risk and did not sustain a hamstring injury, (3) low-risk and sustained a hamstring injury, or (4) low-risk and did not sustain a hamstring injury. The distribution of participants into these 4 groups is the foundation for different metrics (eg, relative risk [RR], odds ratio, true positive, false positive, true negative, false negative, sensitivity, specificity; see Ruddy et al[35] for a detailed description).

The RR (the ratio of the probability of hamstring injury occurring in a "high-risk" group over the probability of hamstring injury occurring in a "low-risk" group) is often reported. It is common to report a specific risk factor resulting in an x-fold (eg, 3-fold) increase in injury risk, which is appealing to researchers and practitioners, alike. However, the robustness of these findings is often overlooked. Relative risk is heavily influenced by a change in injury status of a single participant (eg, a participant deemed "high-risk" moves from the injured group to the noninjured group), especially when sample size and injury occurrence are small.

Several fields and study designs,[11,32] including sports science[52] and sports medicine,[20] have studied the *Fragility Index*. In the context of hamstring injury risk factors, the Fragility Index is the minimum number of participants who would have to change classification from a "high-risk and sustained a hamstring injury" to a "high-risk

OBJECTIVE: To determine the Fragility Index of hamstring injury risk factors, defined as the minimum number of participants who would need to change classification to make a hamstring injury risk factor statistically nonsignificant.

DESIGN: Retrospective secondary data analysis.

METHODS: Studies that investigated 1 or more risk factors for hamstring injury, and presented sufficient data to develop a 2 × 2 contingency table were included. A systematic literature search and reference screening of a recent hamstring injury systematic review were conducted to identify 78 articles. Relative risk and 95% confidence intervals were determined and then systematically recalculated by removing 1 observation from the high-risk injury count and adding it to the high-risk noninjury count. The Fragility Index for a risk factor was the number of observations required to be moved between groups until the relative risk was no longer significant.

RESULTS: The median Fragility Index of all hamstring injury risk factors was 3 (Q1-Q3 = 2-6). The Fragility Index for nonmodifiable risk factors was 3 (Q1-Q3 = 2-6) and 3 (Q1-Q3 = 2-5) for modifiable risk factors. Over 35% of all included hamstring injury risk factors had a Fragility Index of ≤2.

CONCLUSION: Most statistically significant hamstring injury risk factors are fragile associations. The interpretation of significant hamstring injury risk factors should consider a range of statistical metrics, and while the Fragility Index should never be considered in isolation, it is an intuitive measure to help assess the robustness of findings.
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KEY WORDS: *cohort study, Fragility Index, hamstring strain injury, research methodology*

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and did not sustain a hamstring injury" for a risk factor to no longer be statistically significant. The Fragility Index of risk factors for hamstring injury has not been reported.

Therefore, the purpose of this study was to calculate the Fragility Index for risk factors for hamstring injury. By appraising the Fragility Index of hamstring injury risk factors, we aimed to identify risk factors for which replication studies should be a high priority.

METHODS

This was a retrospective analysis of studies that had investigated hamstring injury risk based on 1 or more risk factor/s. A protocol was registered with the International Prospective Register of Systematic Reviews (PROSPERO) database prior to study selection and data extraction (CRD42022360405).

Search Strategy

Studies were obtained from 2 sources. First, the 78 articles included in a recent systematic review focused on hamstring injury risk factors[16] were considered for eligibility. Second, a systematic search replicating the search strategy from an earlier systematic review[16] was performed in Medline, Embase, CINAHL, Cochrane, AMED, SPORTDiscus, AUSPORT, and PEDro from January 1, 2019, to October 1, 2022 (**SUPPLEMENTAL FILE 1**). This was to capture any additional articles that had been published since the systematic search was conducted for the original systematic review.[16] Articles were collated using EndNote (Version X9; Thomson Reuters) and then imported to a web-based software for screening (Covidence; Veritas Health Innovation, Melbourne, Australia).

Study Selection

Studies were deemed eligible based on the following inclusion criteria: (1) investigation of first-time or recurrent hamstring injury risk factors; (2) available data for high- and corresponding low risk populations; (3) presentation of discrete data that we could use to calculate RR (number of injured, uninjured, high-risk, and low-risk participants). Exclusion criteria were (1) discrete injury data for RR calculation not available; (2) any nonstrain hamstring pathologies such as studies examining tendinopathy, nonspecific thigh injuries, hamstring origin avulsions and contusion-type presentations; (3) risk factors or comparator groups that were not hamstring-specific (eg, posterior thigh injury); (4) intervention studies; (5) non-English publications; (6) non–full-text publications.

Following duplicate removal, 2 authors (M.A. and S.G.) independently assessed all studies for eligibility based on the title and abstract and conferred on any conflicts before the same 2 authors completed a full-text screening of the remaining studies, resulting in a final yield for this analysis. An additional author (D.A.O.) was engaged to resolve any conflicts where required.

Risk-of-Bias Assessment

Risk of bias was appraised using the QUality In Prognosis Studies (QUIPS) tool.[18] The QUIPS assesses the quality of prognosis and epidemiological studies.[18] The tool's evaluation is based on 6 domains: participation, attrition, prognostic factor measurement, confounding measurement and account, outcome measurement, and analysis and reporting. Papers included from the original systematic review had their QUIPS scores extracted for the current paper.[16] Two authors (M.A. and S.G.) independently applied the QUIPS tool to the newly included studies from the updated search. Evaluations of each author were compared and any disagreements in assessments using the QUIPS tool were mediated by a third author (D.A.O.), as necessary.

Data Extraction

One author (M.A.) extracted the following variables from the included papers: first author, year of publication, risk factor assessed, number of high-risk individuals, number of injured high-risk individuals, number of low-risk individuals, and number of injured low-risk individuals. For some papers where these data were not readily available but could be calculated based on other information provided, a member of the research team performed these calculations manually. To cross-check these manual calculations, the sum of the injured low-risk and injured high-risk groups was compared to the expected number of hamstring injuries reported in the paper. In instances where the number of calculated hamstring injuries varied by more than ±1 compared to the expected number of hamstring injuries for that variable from that specific study was removed from further analysis.

Risk factors were coded for uniformity (eg, "playing position" was coded to "player position" for consistency across studies) and subgrouped for analysis (eg, "eccentric hamstring torque" and "Nordic strength" were collapsed into "eccentric strength").

Statistical Analysis

All statistical analyses were conducted using the R programming language[30] and the following packages: tidyverse[53] and epitools.[2] For each reported risk factor (ie, variable), RR values were calculated using the following formula:

$$RR = \frac{TP \div (TP + FP)}{FN \div (FN + TN)}$$

where TP = the number of high-risk athletes who sustained a hamstring injury, FP = the number of high-risk athletes who did not sustain a hamstring injury, TN = the number of low-risk athletes who did not sustain a hamstring injury, and FN = the number of low-risk athletes who sustained a hamstring injury.[35]

A RR value and the lower 95% confidence interval (CI) both greater than 1 was considered statistically significant. For all significant variables, the RR value was successively recalculated by removing 1 observation from the TP count (ie, high-risk athlete that sustained hamstring injury) and adding it to the FP count (ie, high-risk athlete did not sustain a hamstring injury) until the RR value was no longer significant (ie, the lower 95% CI was ≤1). The Fragility Index was determined as the total number of observations required to shift from

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the TP count to the FP count until the RR value was no longer significant. All variables were classified as either modifiable or nonmodifiable and were further categorized into subgroups. Medians, Quartile 1 and Quartile 3 (Q1-Q3) of the Fragility Index were calculated.

RESULTS

The primary database search yielded 1762 records. After duplicates were removed, 926 studies were screened and 122 were identified for full-text review. The 78 studies from the original systematic review were automatically included for full-text review. Twelve studies from the updated search and 29 studies from the original review were included after application of eligibility criteria, resulting in a final yield of 41 studies for the analysis.[1,4–10,12–15,17,19,21–23,25–29,31,33,34,36–51,54] The study flow and rationale excluding studies are described in **FIGURE 1**. Approximately one quarter of studies were at low risk of bias across all domains of the QUIPS tool (**SUPPLEMENTAL FILE 2**).

The total number of risk factors investigated in the included studies, the number of significant risk factors, and the median of the Fragility Index for each subgroup are reported in the **TABLE 1**. In total, 262 individual variables were reported; 63 were significant in our analysis. The median Fragility Index across all variables was 3 (Q1-Q3 = 2-6) (**FIGURE 2A**). The distribution of the Fragility Index across modifiable and nonmodifiable risk factors is illustrated in **FIGURE 2B** and **FIGURE 2C**, respectively. The results for each individual risk factor, including the number of TPs, FPs, TNs, and FNs, as well as the original RR value and the corresponding Fragility Index, are included in **SUPPLEMENTAL FILE 3**.

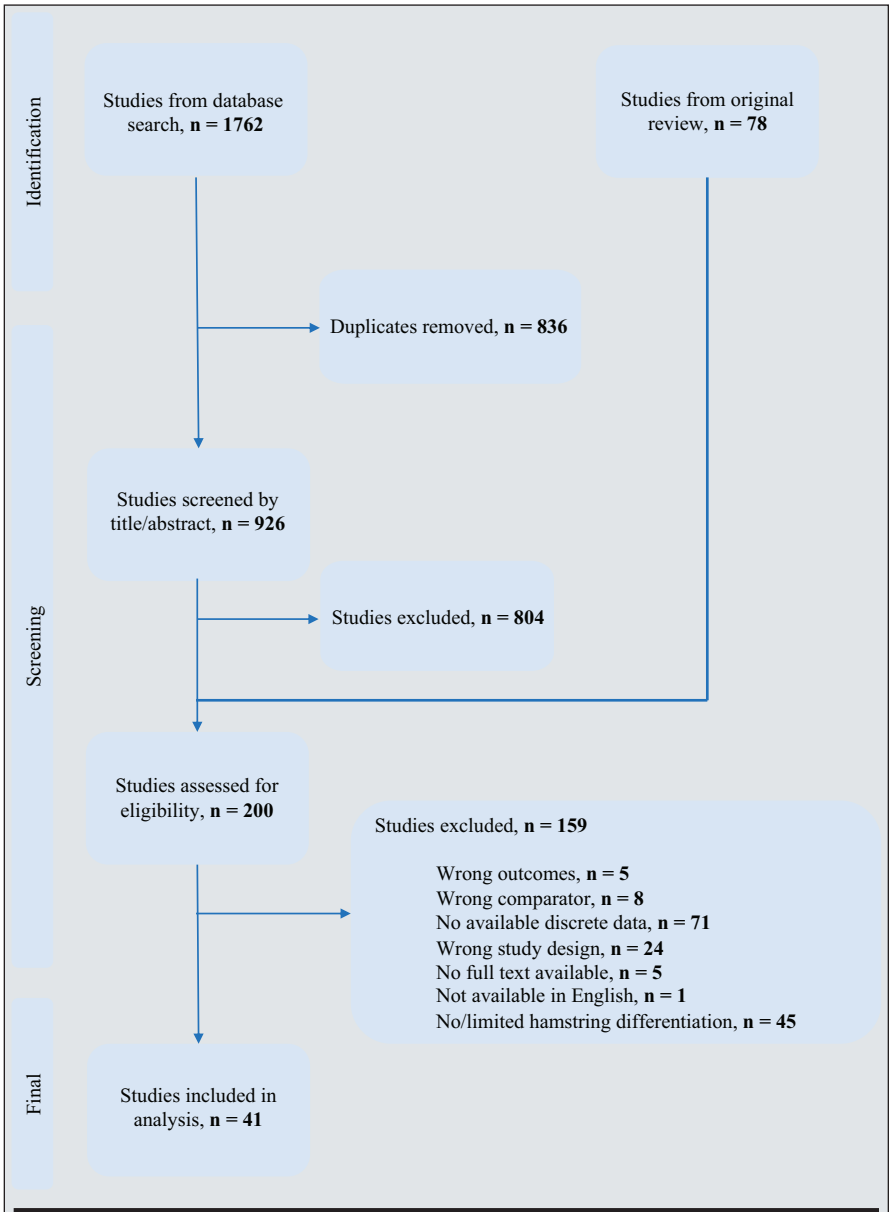

FIGURE 1. Study flowchart (Preferred Reporting Items for Systematic Reviews and Meta-Analyses [PRISMA]) describing the selection of included studies.

DISCUSSION

We calculated the Fragility Index of hamstring injury risk factors, finding that the median (IQR) Fragility Index was 3 (Q1-Q3 = 2-5) for modifiable and 3 (Q1-Q3 = 2-6) for nonmodifiable risk factors. One-third of all significant risk factors for hamstring injury had a Fragility Index of ≤2. Using age as an example, a Fragility Index of 3 indicates that it would require 3 hamstring injuries to not occur in the high-risk (ie, older) group for the risk factor to become nonsignificant.

Previous work has demonstrated similar Fragility Index distributions across domains of sports medicine and arthroscopic surgery,[20] spinal surgery,[11] and critical care.[32] Across these studies,[11,20,32] the median Fragility Index did not exceed 2. Over 80% of sports medicine and arthroscopy randomized controlled trials generated a Fragility Index of ≤3.[20] The 3 most commonly reported hamstring injury risk factors of age, previous hamstring injury, and low eccentric hamstring strength, had a median

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TABLE 1

For Each Risk Factor Type and Subgroup, the Number of Studies Which Have Examined the Risk Factor, the Number of Discrete Times the Variable Was Examined Across Those Studies, the Number of Times the Risk Factor Was Reported as a Significant Risk Factor, and the Median (Quartile 1 [Q1] to Quartile 3 [Q3]) of the Fragility Index Are Reported

Risk Factor Type	Risk Factor Subgroup	Number of Studies Examining the Risk Factor[a]	Number of Discrete Times the Risk Factor Has Been Examined Across All Studies[b]	Times Reported as a Significant Risk Factor Across All Studies	Fragility Index Median (Q1-Q3)
Nonmodifiable	All	60	146	46	3 (2-6)
	Age	8	23	6	3 (2-5.5)
	Anthropometry	4	18	2	2.5 (1.75-3.25)
	Sport or playing position[c]	9	35	12	5.5 (2.75-19.25)
	Ethnicity	3	7	1	2 (2-2)
	Gender	2	2	1	72 (72-72)
	Limb dominance	6	6	2	1 (1-1)
	Pain[d]	2	2	1	5 (5-5)
	Prior hamstring injury	18	21	12	4 (2-7.5)
	Prior hamstring injury time	2	5	3	3 (2-3)
	Prior lower limb injury	6	27	6	2 (2-2.75)
Modifiable	All	17	76	17	3 (2-5)
	Concentric strength	1	1	1	2 (2-2)
	Eccentric strength	6	21	8	3.5 (2.75-4.25)
	Fascicle length	1	6	3	7 (5.5-8)
	Strength imbalance	7	38	4	2 (1-4)
	Length and range of motion	2	10	1	1 (1-1)

[a]*This only includes studies that had the required data available for extraction for the current manuscript.*
[b]*As an individual study may examine the same risk factor in several different ways the number of discrete times the risk factors has been examined will always be greater than or equal to the number of studies examining the risk factor.*
[c]*Defined as different disciplines of sport (eg, football versus track) or within-sport differences in role (eg, goalkeeper versus midfielder).*
[d]*Defined as reported level of pain measured at acute time intervals post hamstring injury.*

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Fragility Index of 3, 4, and 3.5, respectively. Our Fragility Index values provide an additional perspective through which to interpret the clinical implications of research on hamstring injury risk. We also encourage readers to consider the total sample size (of an individual study or the total body of evidence), the prevalence of injury in the given sample, and the number of participants who sustained a hamstring injury. For example, a single study, albeit with a high Fragility Index, should likely be considered preliminary until confirmatory studies are available. Similarly, multiple studies with low Fragility Indexes may be considered weak evidence of an effect, despite their volume (perhaps representing evidence of publication bias). Preliminary evidence may be reasonable for practitioners to act upon where costs are relatively low and/or effect sizes are large. Conversely, strong evidence is required where risks and/or costs are high, and/or effect sizes are small.

Both categorical and continuous risk factors were included in our analysis. While categorical factors (eg, previous injury) fit well into a contingency table, continuous variables require a cut point to distinguish high- and low-risk groups, which is often achieved via the use of a statistical approach (eg, receiver operator characteristic curve). The Fragility Index, as we have implemented it, does not account for the sensitivity and/or specificity of a dichotomized continuous variable, and needs to be considered in context with other metrics to judge the strength and robustness of a hamstring injury risk factor.

We used the cut points provided in each included paper (where available) to dichotomize continuous variables into high- and low-risk groups. While a Fragility Index is presented for a variable across all studies, it is not representative of a single cut point. For example, age has been dichotomized at many different cut points and so the presented Fragility Index of 3 should not be interpretated at a specific age cut point, but instead as a collective Fragility Index for the papers where age was identified as a statistically significant risk factor for future hamstring injury.

Limitations

There may be a systematic bias in only analyzing risk factors that have been previously reported as statistically significant. Similar to how significant risk factors may easily become nonsignificant if 1 or 2 participants classified as high-risk did not get injured,

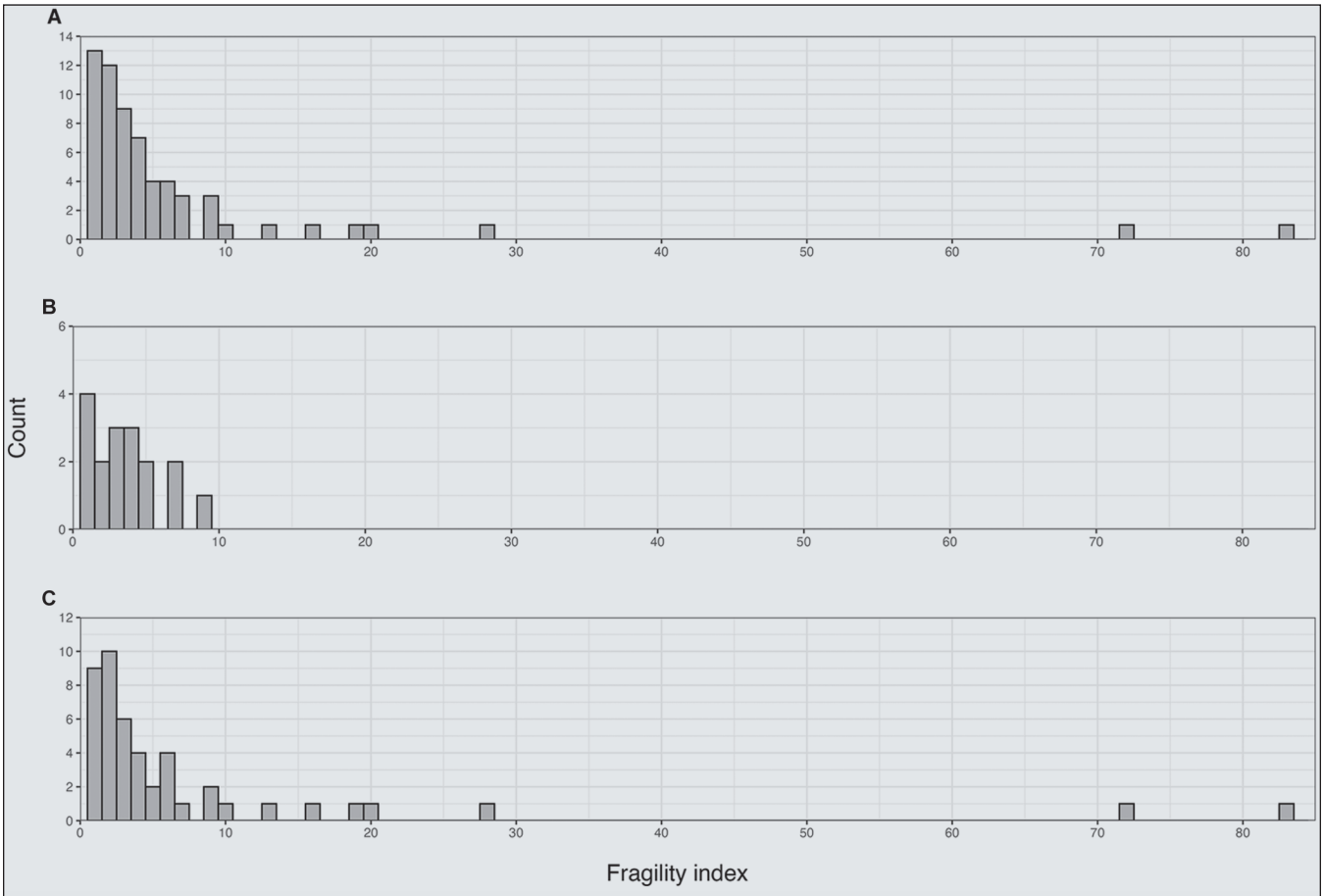

FIGURE 2. Histograms displaying the frequency of Fragility Indices across significant hamstring strain injury risk factors. The Fragility Index was determined as the total number of observations required to shift from the true positive count (the number of high-risk athletes that sustained a hamstring injury) to the false positive count (the number of high-risk athletes that did not sustain a hamstring injury) until the relative risk value was no longer significant. Displayed are the frequencies of Fragility Indices for (A) all risk factors, (B) modifiable risk factors, and (C) nonmodifiable risk factors.

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there may be nonsignificant factors that are 1 or 2 events away from becoming statistically significant. Nonetheless, given the weight placed on significant risk factors by practitioners, we believed it was pertinent to primarily scrutinize these factors.

We were unable to synthesize the entire field of hamstring injury literature because 1 in every 3 potentially eligible studies did not report extractable data required for the analysis. The Fragility Index is limited in its application and interpretability as it is intended to be used in randomized control trials where the sample sizes between the 2 groups are equal. We applied the Fragility Index to observational studies with imbalance between groups, which may have some limitations. Variations to the original Fragility Index have been proposed to account for factors such as covariance and nondichotomous outcomes,[3] and as more alternatives to the traditional Fragility Index become established, these may be more suitable to implement in future work examining observational studies.

Similar to *P* values, the Fragility Index does not indicate the size of an association or effect between a potential risk factor and subsequent outcome (ie, injury occurrence). Different metrics are used to describe the magnitude of an effect or association and measures of variance of these metrics (eg, 95% CIs). The Fragility Index should be considered complementary to existing metrics as it provides additional context to the interpretation of the findings; it should not be considered in isolation.

CONCLUSION

Statistically significant hamstring injury risk factors were often fragile and should be interpreted with caution. ◉

KEY POINTS

FINDINGS: Statistically significant hamstring injury risk factors are commonly fragile and should be interpreted with caution.

IMPLICATIONS: The Fragility Index should be reported in future works to aid in interpretation and synthesis of injury risk factor literature. Practitioners should consider statistical fragility when they appraise the existing body of evidence.
CAUTION: This retrospective analysis study assessed the Fragility Index of hamstring injury risk factors that were statistically significant, subject to systematic and publication bias. The Fragility Index should not be interpreted as a measure of statistical power that is ultimately practice-affirming or -changing, but rather as an additional contextual point for clinicians as they appraise the body of hamstring injury research.

STUDY DETAILS

AUTHOR CONTRIBUTIONS: M.A., S.G., R.W., R.T., D.O., N.M., and J.H. contributed to the study concept. M.A., S.G., B.K., J.R., N.M., and J.H. contributed to the acquisition of data. J.R., B.K., R.W., and D.O. contributed to the analysis and interpretation of data. M.A., S.G., B.K., J.R., and D.O. contributed to the writing/editing of the manuscript. M.A., S.G., B.K., J.R., R.W., R.T., N.M., J.H., and D.O. contributed to the final approval of the manuscript.
DATA SHARING: All data relevant to the study are included in the article or are available as supplementary files.
PATIENT AND PUBLIC INVOLVEMENT: Not applicable.

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